

2.8 SUBSPACES OF $\mathbb{R}^n$

Definition: A subspace of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

- The zero vector is in H .
- For each u and v in H , the sum $u+v$ is in H .
- For each u in H and each scalar c , the vector cu is in H .

In words, a subspace is closed under addition and scalar multiplication.

Example 1: If v_1 and v_2 are in $\mathbb{R}^n$ and $H = \text{Span}\{v_1, v_2\}$, then H is a subspace of $\mathbb{R}^n$. Verify it.

$0 = 0v_1 + 0v_2$. $\therefore 0$ is a linear combination of v_1 and v_2 . By Defn., $0 \in \text{Span}\{v_1, v_2\}$. $\therefore 0 \in H$.

Let $s, t \in H$. By Defn. of Span, s and t are linear combinations of v_1, v_2 .

$$\begin{aligned} \text{By Defn, } \exists \text{ scalars } a_1, a_2, b_1, b_2 \text{ s.t. } & s = a_1v_1 + b_1v_2 \\ & t = a_2v_1 + b_2v_2 \end{aligned}$$